Risk-Aware and Collision-Preventive Cooperative Fleet Cruise Control Based on Vehicular Sensor Networks

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***Abstract*—In this article, we propose a cooperative fleet cruise control (CFCC) framework for collision avoidance and lane changing based on vehicular sensor networks. The proposed CFCC framework is designed for fleets to reduce the recovery time to get the intended speed back after sudden braking, which can improve the driving efficiency of fleets. In addition, CFCC can determine the critical safety spaces of car following and lane changing for fleet vehicles to avoid the accident of rear-end and side collisions due to emergency braking. Furthermore, fleet lane changing can be coordinated between fleet vehicles in the original lane and neighboring vehicles in the target lane for ensuring safety and improving efficiency. According to our review of relevant research, CFCC is the first fleet cruise control system to provide the following features: 1) extending the car following risk of individual vehicles to the fleet following risk of fleet vehicles; 2) minimizing the acceleration and deceleration of fleet vehicles to reduce the speed recovery time; and 3) coordinating individual cars and fleet vehicles to provide the sufficient lane changing space as soon as possible. Simulation results show that our framework outperforms existing car following and lane changing methods and can significantly improve the driving safety and recovery efficiency of fleets as emergency events occur.**

***Index Terms*—Cooperative collision avoidance, fleet cruise control, intelligent transportation system, vehicle-to-vehicle communication, vehicular sensor network (VSN).**

I. Introduction

THE RECENT development of embedding sensor and vehicular communication technologies has made vehicular sensor networks (VSNs) realizable. VSNs achieve the advantages of both vehicular ad hoc networks (VANETs) and wireless sensor networks (WSNs). In VSNs, sensor devices are equipped on vehicles or deployed on roads to provide the sensing capability for traffic and environmental information [1]. Novel VSN-based applications and services have been developed for driving efficiency and traffic safety, including cooperative lane positioning [2], cooperative collision avoidance [3], cooperative traffic control [4], cooperative metropolitan localization [5], etc.

For driving efficiency, traffic flow stability and string stability are significantly influenced by vehicle acceleration/deceleration behavior [6]. The excess accelerating and braking behavior may cause fuel wastage and traffic congestion [7], [8]. The study of improving vehicle fuel economy [9] shows that the additional fuel consumption between the most inefficient and efficient drivers is nearly 70%. In the U.K., the cost caused by traffic congestion is more than 30 billion British pounds and the average time spent by London drivers in traffic congestion is more than 20% in 2016 [10]. In the U.S., the total cost of traffic congestion for drivers is nearly 305 billion U.S. dollars and each driver in Los Angeles spends more than 100 h in traffic congestion during peak time periods in 2017 [11]. In particular, the cost of traffic congestion in time and fuel wastage can be reduced by alleviating the unnecessary acceleration and deceleration of drivers [12]. On the other hand, the specific lanes can be reserved to improve the efficiency of special transports for time-guaranteed purposes [13], and the travel time of vehicles in each region can be stabilized through the perimeter control for urban traffic networks [14].

For traffic safety, existing techniques for advanced driver assistance systems (ADASs) [15] include adaptive cruise control (ACC) [16], [17], cooperative ACC (CACC) [18], [19], lane changing control (LCC) [20]–[22], etc. In particular, the driving safety can be further enhanced by integrating ADAS with car-following behavior [23] and lane changing models [24]. The driving speed of a vehicle is automatically controlled by ACC to maintain a safety following distance from its immediate front vehicle by measuring their distance and relative speed, whereas the automatic speed control is further extended by CACC to use real-time information received from other vehicles and/or roadside units through vehicular communications. The available space in the target lane is ensured by LCC to perform lane changing without collisions. In particular, the driver's intention to change the lane could be recognized through brain–machine interfaces to provide the suitable and timely assistance of lane changing [25].

In ACC systems, Luo *et al.* [16] exploited the predictive driving information provided by ACC for energy management in hybrid electric vehicles (HEVs). The ACC system for intelligent HEVs (i-HEV ACC) is developed to coordinate

Manuscript received January 30, 2020; revised June 15, 2020; accepted December 31, 2020. Date of publication January 27, 2021; date of current version December 17, 2021. This work was supported in part by the Ministry of Science and Technology, Taiwan, under Grant 106-2221-E-035-019-MY3 and Grant 109-2221-E-035-062-MY3. This article was recommended by Associate Editor G. Wen. *(Corresponding author: Lien-Wu Chen.)*

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Color versions of one or more figures in this article are available at https://doi.org/10.1109/TSMC.2021.3050099.

Digital Object Identifier 10.1109/TSMC.2021.3050099

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TABLE I
COMPARISON OF PRIOR WORKS [16]–[22] AND OUR CFCC FRAMEWORK

Features	Adaptive Cruise Control	Cooperative Adaptive Cruise Control	Collision Risk Estimation	Individual Lane Changing	Fleet Lane Changing
i-HEV ACC [16]	√				
DSSM ACC [17]	√		√		
TPCC [18]	√	√	√		
TPAC [19]	√	√			
VSL LCC [20]				√	
SMPC [21]	√		√	√	
CDLC [22]	√	√		√	
Our CFCC Framework	√	√	√	√	√

velocity control for both traffic safety and fuel consumption. Tak *et al.* [17] proposed the deceleration-based surrogate safety measure (DSSM) as a safety indicator for rear-end collision risk estimation during human driving. The DSSM-based ACC can alleviate the potential rear-end collision of a vehicle caused by the unexpected driving behavior of its immediate front vehicle.

In CACC systems, Tak *et al.* [18] further extended the DSSM-based ACC to traffic predictive cruise control (TPCC) for improving traffic flow operation and vehicle fuel efficiency. TPCC can reduce the effects of the shockwave from downstream traffic and alleviate the unnecessary deceleration and acceleration to obtain a higher level of traffic stability. Tsugawa *et al.* [19] discussed truck platooning automation control (TPAC) techniques using sensing technologies, such as computer vision, radar/lidar, satellite localization, vehicular communications, etc. TPAC can achieve energy saving and enhance transportation capacity, where the unmanned operation of following trucks can reduce personnel cost.

In LCC systems, Zhang and Ioannou [20] developed a variable speed limit (VSL)-based LCC scheme to improve travel time in traffic incidents. The VSL-based LCC scheme can reduce the effect of the capacity drop by recommending lane changing actions upstream the incident location or traffic bottleneck. Suh *et al.* [21] designed stochastic model-predictive control (SMPC) for automated driving to plan lane change motion with probabilistic and deterministic prediction. SMPC uses predicted time-to-collision and safety distance to guarantee safe lane changing through the lane changing risk estimated. He *et al.* [22] proposed a new model of cooperative driving and lane changing (CDLC) under the vicinity of traffic signals with vehicular communications. The CDLC model can reduce the stopping frequency and travel time of vehicles by planning vehicle trajectories in the vicinity of traffic signals in advance. In addition, CDLC can improve safety and comfort during the lane changing process by planning the cooperative strategy of lane changing based on the relationship between the host vehicle in the original lane and surrounding vehicles in the target lane.

On the other hand, collision risk assessment has been investigated in existing works [26]–[28]. Through VSN technologies, preventing rear-end collisions among vehicles can be achieved by utilizing car following risk (CFR) estimated based on the instant vehicle state and accurate driving information. However, the current collision risk assessment methods are designed for individual vehicles, which do not consider the specific property of fleet vehicles. In addition, the safe lane changing of fleet vehicles is not addressed in existing ADAS techniques (designed for individual vehicles). Furthermore, the efficient car following and lane changing of fleet vehicles are not explored in the aforementioned ACC, CACC, and LCC systems to minimize their acceleration and deceleration for fuel saving and congestion relieving.

In this work, we design a cooperative fleet cruise control (CFCC) framework for collision avoidance and lane changing based on VSNs. The designed CFCC framework can improve the driving efficiency of a fleet by reducing the recovery time to get the intended speed back after sudden braking. In addition, CFCC can determine the critical safety spaces of car following and lane changing for the fleet to avoid the accident of rear-end and side collisions due to emergency braking. In particular, CFCC estimates CFR and fleet following risk (FFR) between individual vehicles and between fleet vehicles, respectively, for both cooperative collision avoidance and vehicle acceleration/deceleration minimization. Furthermore, fleet lane changing can be coordinated among fleet vehicles in the original lane and neighboring vehicles in the target lane for alleviating the unnecessary acceleration/deceleration of drivers and improving traffic flow stability and string stability.

Table I shows a comparison of the functions provided by existing schemes with ours. Our CFCC framework offers the most complete solution to cooperative fleet car following and lane changing. According to our review of relevant research, CFCC is the first fleet cruise control system to provide the following features: 1) the CFR of individual vehicles is estimated and further extended to the FFR of fleet vehicles for cooperative collision avoidance; 2) the acceleration and deceleration of fleet vehicles are minimized through the extended FFR for driving efficiency improvement; and 3) the lane change space needed for a fleet is coordinated among individual vehicles and fleet vehicles for rapid space provision.

The contributions of our framework are fourfold. First, the CFR of a vehicle with its immediate front vehicle is estimated according to the driver's reaction time, following distance, driving speed, and maximum deceleration of vehicles. Second, the FFR of fleet vehicles extended from CFR of individual vehicles is used to coordinate the velocity control of vehicles in a fleet. Third, the necessary acceleration and deceleration periods of individual vehicles in the target lane are derived for the lane changing of fleet vehicles in the original lane. Finally, the feasible deceleration periods of fleet vehicles in

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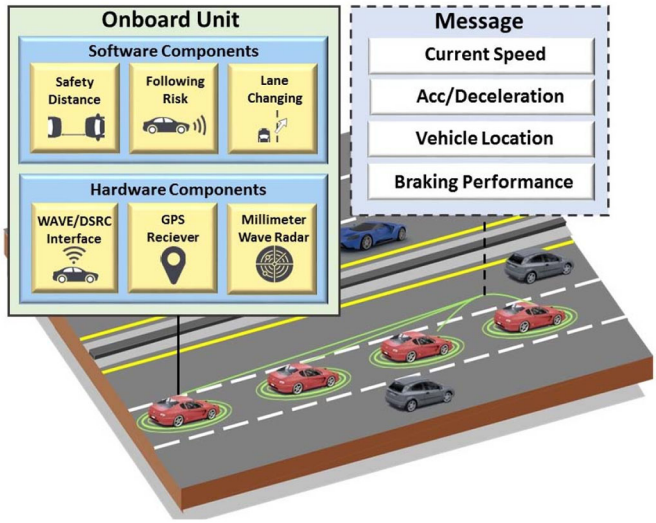

Fig. 1. System architecture of CFCC.

the original lane and the necessary acceleration/deceleration periods of individual vehicles in the target lane are cooperatively determined for both fleet braking and lane changing. Furthermore, simulation results show that our framework outperforms existing methods and can significantly improve the driving safety and recovery efficiency of fleets as emergency occurs.

The remainder of this article is organized as follows. Section II defines our cooperative cruise control problem for fleets based on VSN. Section III presents our framework to solve this problem. Simulation results are discussed in Section IV. Finally, Section V concludes this article.

II. System Model

Fig. 1 shows the system architecture of CFCC. Fleet vehicles and nonfleet vehicles (i.e., individual vehicles) driving on the roads form VSNs, which can communicate with each other via vehicle-to-vehicle (V2V) communications using IEEE 802.11p-based [30] wireless access in vehicular environments/dedicated short range communications (WAVE/DSRC) [31] network interfaces. The specific wildcard BSSID is used in all WAVE/DSRC network interfaces operating in the same channel by default. Thus, the required connection setup time between vehicles is short because two encountering vehicles do not need to perform association and authentication procedures.

According to the U.S. Federal Communication Commission, the 75-MHz spectrum of WAVE/DSRC at 5.9 GHz consists of six service channels and one control channel, where safety-related messages of CFCC are allowed to be transmitted in either the control channel (i.e., channel 178) or the dedicated public safety V2V service channel (i.e., channel 172). The WAVE/DSRC network interface is periodically switching between the control channel and service channel every 50 ms (i.e., broadcast frequency is 10 Hz), where the transmission delay and packet loss of CFCC safety messages can be minimized by assigning the highest priority [via enhanced distributed channel access (EDCA) adopted in IEEE 802.11p] in the dedicated channel. The vehicle state information, including the driving speed, acceleration/deceleration, current location, and accelerating/braking performance (i.e., maximum acceleration/deceleration speed), can be efficiently broadcasted to nearby vehicles in periodical/event-driven messages by using the WAVE/DSRC network interface.

For fleet vehicles, an onboard unit (OBU) is installed inside a car to receive vehicle state information from neighboring vehicles in the communication ranging. The OBU consists of a WAVE/DSRC network interface for V2V communications, a GPS receiver for location measurement, and a millimeter wave radar for following distance estimation. In CFCC, the safety distance that has to be kept between two consecutive fleet vehicles is calculated and the following risk of a vehicle with its immediate front vehicle in the same fleet is estimated in a real-time manner. In addition, the sufficient space for fleet lane changing (from the original lane to the target lane) can be provided by adjusting the speeds of neighboring vehicles in the target lane with the deceleration coordination of the fleet in the original lane if necessary.

For the cooperative cruise control of fleet vehicles, we formulate the problem as follows. First, when a vehicle is driving in front of another vehicle in the same lane, the following distance between them must be sufficient to keep the CFR low for avoiding their rear-end collision. Second, the CFR of individual vehicles is extended to the FFR of fleet vehicles for cooperative collision avoidance. Third, the speeds of neighboring vehicles in the target lane are controlled to obtain the available space for the lane changing of fleet vehicles in the original lane. Finally, the speeds of fleet vehicles and neighboring vehicles are adjusted cooperatively to reduce the recovery time of getting the intended fleet speed back. Our goal is to design efficient mechanisms to accurately estimate the following risk for fleet cruise safety and quickly provide the sufficient space for fleet lane changing by addressing the following four research issues.

1) *Following Risk Estimation:* How do we accurately estimate the CFR of a vehicle with its immediate front vehicle according to the driver's reaction time of the vehicle, the distance between the vehicle and its immediate front vehicle, and their current speeds and maximum deceleration?
2) *Velocity Control Coordination:* How do we efficiently coordinate the velocity control of vehicles in a fleet for rear-end collision avoidance based on both the CFR of individual vehicles and the extended FFR of fleet vehicles?
3) *Lane Space Adjustment:* How do we precisely derive the necessary acceleration and deceleration periods of neighboring vehicles in the target lane to provide a sufficient space for the lane changing of a fleet in the original lane on a basis of the numbers and locations of neighboring vehicles and fleet vehicles?
4) *Cooperative Lane Changing:* How do we cooperatively determine the feasible deceleration period of a fleet in the original lane and the necessary acecleration/deceleration periods of neighboring vehicles in the

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TABLE II
SUMMARY OF NOTATIONS

Notation	Meaning
b_i	Length of vehicle C_i
(X_i, Y_i)	Location of vehicle C_i
S_i	Speed of vehicle C_i
a_i^{max}	Maximum deceleration of vehicle C_i
$d_{i-1,i}$	Following distance between vehicles C_{i-1} and C_i
τ_i	Emergency braking time to completely stop (= S_i/a_i^{max})
δ	Average driver's reaction time
R	Maximum communication range
CFR_i	Car following risk of individual vehicle C_i
FFR_i	Fleet following risk of fleet vehicle C_i
C_{j-k}	The k-th vehicle in lane j
a_{j-k}^{acc}	Maximum acceleration of vehicle C_{j-k}
t_{j-k}	Acceleration/deceleration period of vehicle C_{j-k}
E_{j-k}	Safety following distance of vehicle C_{j-k}

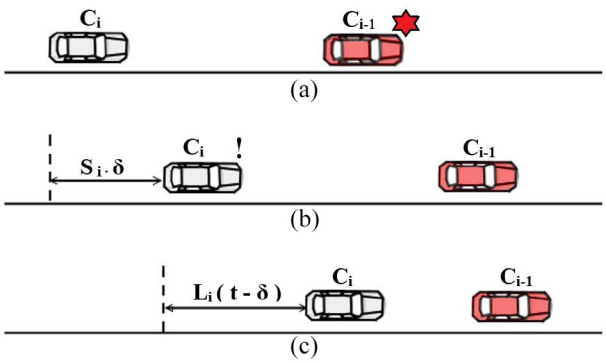

Fig. 2. From (a) seeing the braking light of C_{i-1}, different distances traveled by C_i (b) before and (c) after its emergency brake.

target lane when the fleet has to take a brake before fleet lane changing can be performed?

III. COOPERATIVE FLEET CRUISE CONTROL

In this section, we present the cooperative cruise control framework for fleets, which consists of FFR estimation and fleet lane changing coordination. First, the CFR of individual vehicles is estimated. Second, the estimated CFR of individual vehicles is extended to the FFR of fleet vehicles. Third, the acceleration and deceleration of neighboring vehicles are coordinated in the target lane for fleet lane changing. Finally, the deceleration of fleet vehicles in the original lane is cooperatively determined with the coordinated acceleration and deceleration of neighboring vehicles in the target lane as necessary. Table II lists the summary of notations used in the following risk estimation and lane changing coordination for fleet vehicles.

A. Fleet Following Risk Estimation

For each vehicle C_i and its immediate front vehicle C_{i-1} located at (X_i, Y_i) and (X_{i-1}, Y_{i-1}) (here the centering point of the front end of a vehicle is used as the location point of the vehicle) in the same lane, respectively, there is no collision occurring between C_i and C_{i-1} if the sufficient following distance $d_{i-1,i}$ is kept by C_i. In other words, when C_{i-1} takes an emergency brake [Fig. 2(a)], the sum of driver's reaction time [Fig. 2(b)] and emergency braking time of C_i [Fig. 2(c)] has to be less than the available maneuvering time (proportional to $d_{i-1,i}$) to avoid the rear-end collision between C_i and C_{i-1}. The following distance $d_{i-1,i}$ between C_i and C_{i-1} is

$$d_{i-1,i} = |X_{i-1} - X_i| - b_{i-1} \tag{1}$$

where b_{i-1} is the vehicle length of C_{i-1}.

Suppose that the maximum emergency deceleration of C_i is $a_i^{\max}$ and the time for C_i to decelerate its current speed S_i to 0 km/h is $\tau_i = (S_i/a_i^{\max})$. Let $L_i(t)$ be the distance traveled by C_i from its emergency brake (at time 0) to completely stopping (at time t), which can be calculated as

$$L_i(t) = S_i \cdot \tau_i - \frac{1}{2} \cdot a_i^{\max} \cdot \tau_i^2. \tag{2}$$

To obtain the sufficient available maneuvering time during the emergency brake of C_{i-1} [3], the first safety condition has to be satisfied before C_i starts braking

$$L_{i-1}(t) + d_{i-1,i} > S_i \cdot t \quad \forall\ 0 < t \leq \delta \tag{3}$$

where δ is the driver's reaction time of C_i from seeing the braking light of C_{i-1} to starting its emergency brake and the average value of δ is 1.5 s [29]. The first safety condition guarantees that there is no rear-end collision occurring in δ because C_i will keep the same speed S_i until the end of δ after C_{i-1} starts braking. That is, $S_i \times t$ has to be less than the sum of $L_{i-1}(t)$ and $d_{i-1,i}$ within δ to avoid the rear-end collision between C_i and C_{i-1}.

Similarly, the second safety condition has to be satisfied after C_i starts braking

$$L_{i-1}(t) + d_{i-1,i} > S_i \cdot \delta + L_i(t - \delta) \quad \forall\ \delta < t \leq \delta + \tau_i. \tag{4}$$

The second safety condition guarantees that there is no rear-end collision occurring from δ to $\delta + \tau_i$ when C_i starts braking after δ until C_i completely stops. That is, the sum of $S_i \times \delta$ and $L_i(t-\delta)$ has to be less than the sum of $L_{i-1}(t)$ and $d_{i-1,i}$ until S_i becomes 0 km/h to avoid the rear-end collision between C_i and C_{i-1}. If any t within $\delta + \tau_i$ violates the first or second safety condition, the rear-end collision occurs during C_{i-1} takes the emergency brake. From (2)–(4), we can determine the deceleration a_i that C_i has to take at least to avoid the rear-end collision with C_{i-1} as

$$\begin{aligned} &S_{i-1} \cdot \tau_{i-1} - \frac{1}{2} \cdot a_{i-1}^{\max} \cdot \tau_{i-1}^2 + d_{i-1,i} \\ &\quad > S_i \cdot \left(\frac{S_i}{a_i} + \delta\right) - \frac{1}{2} \cdot a_i \cdot \left(\frac{S_i}{a_i}\right)^2. \end{aligned} \tag{5}$$

From (5), a_i can be derived as

$$a_i = \frac{S_i^2 \cdot a_{i-1}^{\max}}{S_{i-1}^2 + 2 \cdot a_{i-1}^{\max} \cdot (d_{i-1,i} - S_i \cdot \delta)}. \tag{6}$$

So the CFR of C_i can be estimated as

$$\text{CFR}_i = \frac{a_i}{a_i^{\max}}. \tag{7}$$

Through CFR_i, it is efficient to measure the safety level of C_i and determine whether C_i is safe or not as an emergency event occurs. If CFR_i is smaller than or equal to 1, C_i can

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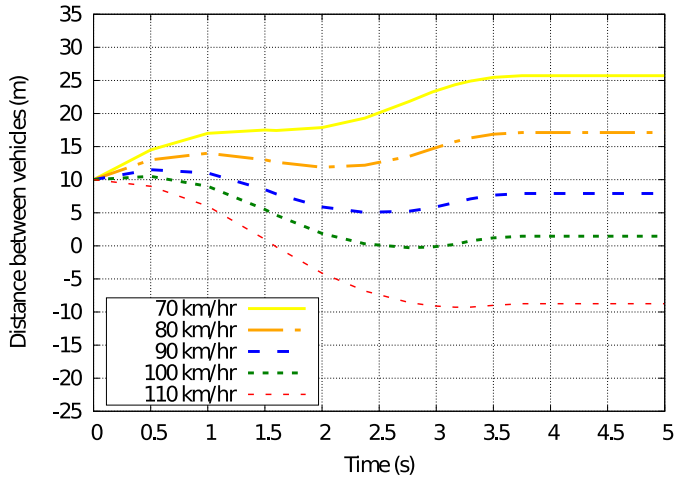

Fig. 3. Distance variations between C_i and C_{i-1} under different driving speeds.

completely stop in time before colliding with C_{i-1} that takes an emergency brake. In contrast, if CFR_i is greater than 1, C_i has to take the deceleration speed larger than its maximum emergency deceleration (which is impossible to achieve) to avoid the rear-end collision when C_{i-1} takes an emergency brake. Therefore, when the value of CFR_i is larger, the risk of a rear-end collision for C_i is higher. In particular, the car following distance between C_i and C_{i-1} has to be increased immediately as CFR_i is greater than 1.

Theoretical safety analysis using CFR_i of C_i is as follows. The road capacity can be increased by CFCC through the smaller following distance gap and shorter reaction time delay. However, the small distance/time gap between the first and following vehicles could improve driving efficiency, but the risk of rear-end collisions might be risen. Distance variations related to driving speeds are analyzed in the theoretical safety of vehicles using CFR. Considering a typical car following scenario with different driving speeds in Fig. 3, which shows distance variations between C_i and C_{i-1} under different driving speeds of C_i as C_{i-1} takes an emergency brake. The initial distance between C_i and C_{i-1} is 10 m, the speed of C_{i-1} is 110 km/h, and the speeds of C_i are 70, 80, 90, 100, and 110 km/h. The emergency brake is taken by C_{i-1} at the 0th second with the maximum deceleration speed of 8 m/s^2. Note that after C_{i-1} takes an emergency brake, the following distance of C_i with 70 km/h is still increasing. This is because the speed of C_{i-1} (i.e., 110 km/h) is much higher than that of C_i. Even after braking for 1.5 s, the speed of C_{i-1} is still higher than 70 km/h.

After the driver's reaction time of 1.5 s, C_i starts braking with the maximum deceleration speed of 14 m/s^2. If the distance between C_i and C_{i-1} is smaller than or equal to 0 at any time point, there is a rear-end collision occurring on C_i with C_{i-1} (i.e., as C_i is driving at the speeds of 100 and 110 km/h in Fig. 3). Through (6) and (7), the risk values of CFR_i for C_i under the speeds of 70, 80, 90, 100, and 110 km/h are estimated as 0.3932, 0.5198, 0.7960, 1.1795, and 1.5126, respectively. It can be analyzed that the car following safety of C_i can be increased by reducing the risk value of CFR_i, which has to keep CFR_i smaller than 1 at all time points.

Next, we further extend the CFR of individual vehicles to the FFR of fleet vehicles. The first vehicle C_1 (in the CFCC mode) in a fleet can calculate FFR_1 with its immediate front vehicle A (not in the CFCC mode) based on (6) and (7) (i.e., $\mathrm{FFR}_1 = \mathrm{CFR}_1$), whereas the second vehicle C_2 (in the CFCC mode) in the same fleet can calculate the FFR_2 through the following equations:

$$S_1 \cdot \tau_1 - \frac{1}{2} \cdot a_1^{\max} \cdot \tau_1^2 + d_{1,2} = S_2 \cdot \left(\frac{S_2}{a_2}\right) - \frac{1}{2} \cdot a_2 \cdot \left(\frac{S_2}{a_2}\right)^2 \tag{8}$$

$$a_2 = \frac{S_2^2 \cdot a_1^{\max}}{S_1^2 + 2 \cdot a_1^{\max} \cdot d_{1,2}} \tag{9}$$

$$\mathrm{FFR}_2 = \frac{a_2}{a_2^{\max}}. \tag{10}$$

Compared with (5), the driver's reaction time δ is removed in (8) because the braking message from C_1 is immediately broadcasted to C_2 for performing an instant brake with deceleration a_2 through V2V communications in the CFCC mode when C_1 takes an emergency brake with maximum deceleration $a_1^{\max}$. In the CFCC mode, C_2 automatically starts braking once it receives the braking message from C_1. Through (8), deceleration a_2 and FFR_2 can be derived in (9) and (10), respectively.

Similarly, the third vehicle C_3 (in the CFCC mode) in the same fleet can determine its deceleration a_3 (and FFR_3) as follows by calculating a_2 first:

$$a_3 = \frac{S_3^2 \cdot a_2}{S_2^2 + 2 \cdot a_2 \cdot d_{2,3}}. \tag{11}$$

For arbitrary fleet vehicle C_i in the CFCC mode, its deceleration a_i (and FFR_i) can be determined as follows by calculating $a_2, a_3, \ldots,$ and a_{i-1} first:

$$a_i = \frac{S_i^2 \cdot a_{i-1}}{S_{i-1}^2 + 2 \cdot a_{i-1} \cdot d_{i-1,i}} \tag{12}$$

where $a_1 = a_1^{\max}$. Note that in the case of V2V communication delay, $d_{i-1,i}$ in (12) can be replaced by $d_{i-1,i} - (S_i \times T_d)$ to further ensure driving safety, where T_d is the delay time of packet transmissions.

Theoretical recovery time analysis using different following risks is as follows. Speed variations and recovery times are analyzed in the theoretical recovery times of vehicles using CFR and FFR. Considering different fleet vehicle scenarios with the same driving speed in Figs. 4 and 5, which show speed variations using CFR and FFR in a fleet with eight vehicles $C_1, C_2, \ldots,$ and C_8, respectively, as the vehicle A in front of the fleet takes an emergency brake. The speed and maximum deceleration of all vehicles (i.e., $A, C_1, C_2, \ldots,$ and C_8) are 30 m/s and 8 m/s^2. The front vehicle A takes an emergency brake at the 0th second and get the intended speed back (i.e., 30 m/s) at the 3rd second. It can be observed that the first fleet vehicle C_1 in both Figs. 4 and 5 starts braking at the 1.5th second after the average reaction time. However, the second fleet vehicle C_2 using CFR starts braking at the 3rd

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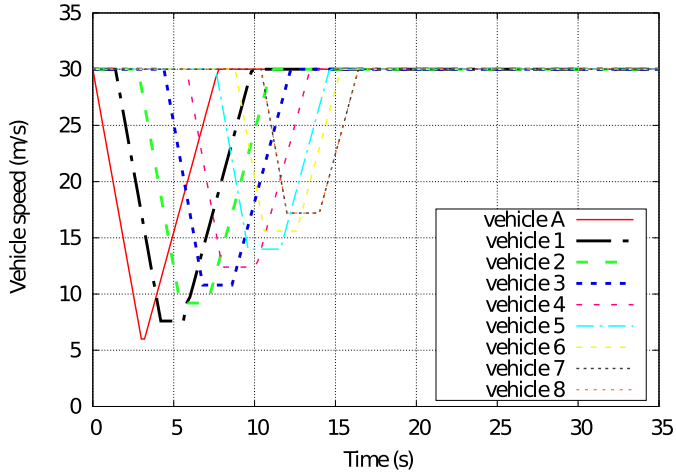

Fig. 4. Speed variations using CFR in a fleet with 8 vehicles.

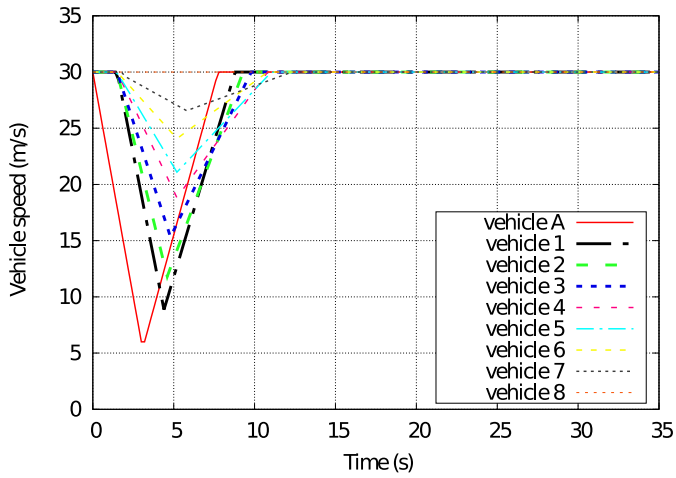

Fig. 5. Speed variations using FFR in a fleet with 8 vehicles.

second, and C_2 using FFR starts braking at the 1.5th second. In particular, C_2, C_3, ..., and C_8 using FFR start braking simultaneously in the CFCC mode, whereas C_2, C_3, ..., and C_8 using CFR start braking sequentially after seeing the braking light of their immediate front vehicles. In addition, C_2, C_3, ..., and C_8 using FFR take a brake with smaller deceleration and get the original speed back more quickly than C_2, C_3, ..., and C_8 using CFR, which can alleviate the influence to traffic flows in the same lane and reduce the fuel wastage of fleet vehicles caused by sudden braking.

B. Fleet Lane Changing Coordination

In this section, we investigate the efficient coordination of lane changing among neighboring vehicles in the target lane for a fleet as there are traffic incidents (e.g., car accident or road obstacle) occurring in the original lane of the fleet. Based on the number and locations of neighboring vehicles in the target lane, four different cases are addressed to obtain the sufficient space in the target lane for fleet lane changing. In addition, the safety following distances have to be kept for the neighboring vehicles in front and back of the fleet after lane changing. In particular, both the CFR of neighboring vehicles

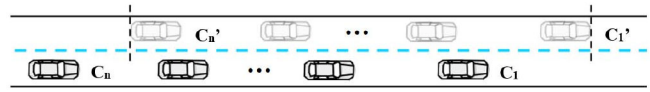

Fig. 6. Lane changing space needed for a fleet with n vehicles.

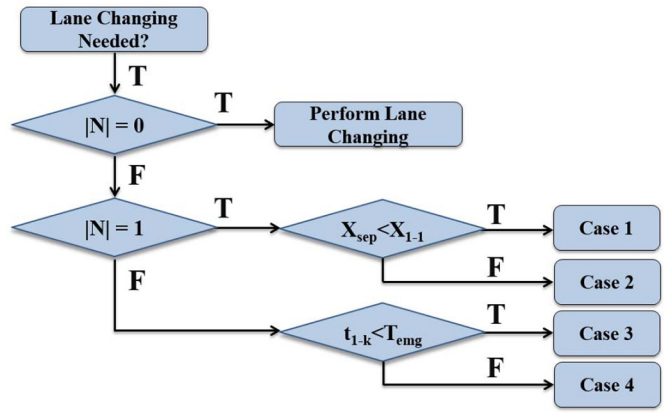

Fig. 7. Flowchart of fleet lane changing in CFCC.

and the FFR of fleet vehicles have to be always smaller than 1 before, during, and after fleet lane changing. As shown in Fig. 6, a fleet with n vehicles C_1, C_2, ..., and C_n are located at (X_1, Y_1), (X_2, Y_2), ..., and (X_n, Y_n) in the original lane, respectively, and the estimated space needed by C_1', C_2', ..., and C_n' after fleet lane changing is the distance between X_1' and X_n' in the target lane.

Fig. 7 shows the flowchart of fleet lane changing in CFCC, where the target and original lanes are lane 1 and lane 2, respectively. When lane changing is needed for a fleet F with n vehicles C_{2-1}, C_{2-2}, ..., and C_{2-n} in lane 2 (due to traffic congestion or road incidents in the same lane), the first fleet vehicle C_{2-1} constructs a set N of F's neighboring vehicles in lane 1 (by collecting the neighboring vehicle information of each fleet vehicle in F through multihop V2V communications), where all neighboring vehicles of F are located between $X_{2-n}-R$ and $X_{2-1}+R$ and the maximum communication range of each vehicle is R.

Next, C_{2-1} further checks whether the future location (X_{1-i}', Y_{1-i}') of each neighboring vehicle C_{1-i} in N after fleet lane changing is in the estimated space S_F needed by C_{2-1}', C_{2-2}', ..., and C_{2-n}' of F as

$$X_{2-n}' - E_{1-i} < X_{1-i}' < X_{2-1}' + E_{2-1} \tag{13}$$

where

$$E_{j-k} = S_{j-k} \cdot (\delta + \tau_{j-k}) - \frac{1}{2} \cdot a_{j-k}^{\max} \cdot \tau_{j-k}^2. \tag{14}$$

In (13), E_{j-k} represents the maximum safety following distance of C_{j-k} to avoid the rear-end collision with its immediate front vehicle. All neighboring vehicles that are not satisfying (13) will be removed from N. If there is no satisfied neighboring vehicle in N (i.e., $|N| = 0$), it implies that there is already a sufficient space of S_F without any influenced neighboring vehicle and fleet lane changing can be done by F immediately (i.e., perform lane changing in Fig. 7).

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On the other hand, if there is only one satisfied neighboring vehicle C_{1-1} in N (i.e., $|N| = 1$), it implies that C_{1-1} has to accelerate (i.e., case 1 in Fig. 7) or decelerate (i.e., case 2 in Fig. 7) to leave S_F (depending on whether the current lateral location X_{1-1} of C_{1-1} is greater than X_{sep} or not) and fleet lane changing can be done by F after C_{1-1} moves outside S_F. X_{sep} is the separate location such that the time t_{1-1}^{acc} of accelerating at X_{sep} to move outside S_F is equal to the time t_{1-1}^{dec} of decelerating at X_{sep} to move outside S_F (spent by C_{1-1}), which can be found by solving the following simultaneous equations:

$$X_{2-1} + E_{2-1} - X_{\text{sep}} = (S_{1-1} - S_{2-1}) \cdot t_{1-1}^{\text{acc}} + \frac{1}{2} \cdot a_{1-1}^{\text{acc}} \cdot (t_{1-1}^{\text{acc}})^2 \tag{15}$$

and

$$X_{\text{sep}} + E_{1-1} - X_{2-n} = (S_{2-n} - S_{1-1}) \cdot t_{1-1}^{\text{dec}} + \frac{1}{2} \cdot a_{1-1}^{\max} \cdot (t_{1-1}^{\text{dec}})^2 \tag{16}$$

where $t_{1-1}^{\text{acc}} = t_{1-1}^{\text{dec}}$. Note that if C_{1-1} is accelerating to the maximum speed limit $S_{\max}$ of the road before moving outside S_F, the right-hand side of (15) has to be replaced by

$$(S_{1-1} - S_{2-1}) \cdot \tau_{1-1}^{\text{acc}} + \frac{1}{2} \cdot a_{1-1}^{\text{acc}} \cdot (\tau_{1-1}^{\text{acc}})^2 + (S_{1-1}^{\max} - S_{2-1}) \cdot (t_{1-1}^{\text{acc}} - \tau_{1-1}^{\text{acc}}) \tag{17}$$

where

$$\tau_{1-1}^{\text{acc}} = \frac{S_{\max} - S_{1-1}}{a_{1-1}^{\text{acc}}}. \tag{18}$$

Similarly, if C_{1-1} is decelerating to completely stopping before moving outside S_F, the right-hand side of (16) has to be replaced by

$$(S_{2-n} - S_{1-1}) \cdot \tau_{1-1}^{\text{dec}} + \frac{1}{2} \cdot a_{1-1}^{\max} \cdot (\tau_{1-1}^{\text{dec}})^2 + S_{2-n} \cdot (t_{1-1}^{\text{dec}} - \tau_{1-1}^{\text{dec}}) \tag{19}$$

where

$$\tau_{1-1}^{\text{dec}} = \frac{S_{1-1}}{a_{1-1}^{\max}}. \tag{20}$$

Moreover, if there are more than one satisfied neighboring vehicles $C_{1-1}, C_{1-2}, \ldots$, and C_{1-m} in N (i.e., $|N| = m > 1$), it implies that $C_{1-1}, C_{1-2}, \ldots$, and $C_{1-(k-1)}$ have to accelerate to leave S_F whereas $C_{1-k}, C_{1-(k+1)}, \ldots$, and C_{1-m} have to decelerate to leave S_F (depending on the time t_{1-k} to move outside S_F spent by C_{1-k}) and fleet lane changing can be done by F after $C_{1-1}, C_{1-2}, \ldots$, and C_{1-m} move outside S_F (i.e., case 3 in Fig. 7), where $X_{1-k} \le X_{\text{sep}}$ and $X_{1-(k-1)} > X_{\text{sep}}$. T_{emg} is the remaining time that C_{2-1} can keep the same speed S_{2-1} without any braking before C_{2-1} arrives at the location X_{emg} of the emergency incident occurring in the same lane, which can be calculated as

$$T_{\text{emg}} = \frac{X_{\text{emg}} - X_{2-1}}{S_{2-1}}. \tag{21}$$

If $t_{1-k} > T_{\text{emg}}$, it implies that C_{2-1} has to take a brake before fleet lane changing can be done by F. In other words, $C_{1-1}, C_{1-2}, \ldots$, and C_{1-m} cannot move outside S_F in time for the fleet lane changing of F (i.e., case 4 in Fig. 7). Note that if $t_{1-k} < t_{1-(k-1)}$ (i.e., the acceleration period of $C_{1-(k-1)}$ is greater than the deceleration period of C_{1-k}) or $C_{1-(k-1)} = C_{1-m}$ (i.e., all satisfied neighboring vehicles must accelerate to leave S_F), $t_{1-(k-1)} > T_{\text{emg}}$ is checked instead of $t_{1-k} > T_{\text{emg}}$. In addition, if either $t_{1-1}^{\text{acc}} > T_{\text{emg}}$ (in case 1) or $t_{1-1}^{\text{dec}} > T_{\text{emg}}$ (in case 2), case 4 is used instead to perform fleet lane changing.

Next, we discuss how to accurately coordinate one or more satisfied neighboring vehicles in N to move outside S_F for F consisting of $C_{2-1}, C_{2-2}, \ldots$, and C_{2-n} in each case.

Case 1: When C_{1-1} has to accelerate to leave S_F, it uses its maximum acceleration speed a_{1-1}^{acc} to move outside S_F for the period of t_{1-1} that can be estimated as

$$X_{2-1} + E_{2-1} - X_{1-1} = (S_{1-1} - S_{2-1}) \cdot t_{1-1} + \frac{1}{2} \cdot a_{1-1}^{\text{acc}} \cdot (t_{1-1})^2 \tag{22}$$

where E_{2-1} is the safety following distance of C_{2-1} and $X_{2-1} - X_{1-1}$ is the distance between C_{2-1} and C_{1-1}.

Case 2: When C_{1-1} has to decelerate to leave S_F, it uses its maximum deceleration speed $a_{1-1}^{\max}$ to move outside S_F for the period of t_{1-1} that can be estimated as

$$X_{1-1} + E_{1-1} - X_{2-n} = (S_{2-n} - S_{1-1}) \cdot t_{1-1} + \frac{1}{2} \cdot a_{1-1}^{\max} \cdot (t_{1-1})^2 \tag{23}$$

where E_{1-1} is the safety following distance of C_{1-1} and $X_{1-1} - X_{2-n}$ is the distance between C_{1-1} and C_{2-n}.

Case 3: When $C_{1-1}, C_{1-2}, \ldots$, and $C_{1-(k-1)}$ have to accelerate to leave S_F, they use their maximum acceleration speeds $a_{1-1}^{\text{acc}}, a_{1-2}^{\text{acc}}, \ldots$, and $a_{1-(k-1)}^{\text{acc}}$ to move outside S_F for the periods of $t_{1-1}, t_{1-2}, \ldots$, and $t_{1-(k-1)}$, respectively. The acceleration periods $t_{1-1}, t_{1-2}, \ldots$, and $t_{1-(k-1)}$ can be estimated as

$$X_{2-1} + E_{2-1} - X_{1-(k-1)} = (S_{1-(k-1)} - S_{2-1}) \cdot t_{1-(k-1)} + \frac{1}{2} \cdot a_{1-(k-1)}^{\text{acc}} \cdot (t_{1-(k-1)})^2 \tag{24}$$

$$X_{2-1} + E_{2-1} + E_{1-(k-1)} - X_{1-(k-2)} = (S_{1-(k-2)} - S_{2-1}) \cdot t_{1-(k-2)} + \frac{1}{2} \cdot a_{1-(k-2)}^{\text{acc}} \cdot (t_{1-(k-2)})^2 \tag{25}$$

$$\vdots$$

$$X_{2-1} + E_{2-1} + \sum_{i=2}^{k-1} E_{1-i} - X_{1-1} = (S_{1-1} - S_{2-1}) \cdot t_{1-1} + \frac{1}{2} \cdot a_{1-1}^{\text{acc}} \cdot (t_{1-1})^2. \tag{26}$$

Note that if there is no sufficient space in lane 1 for the acceleration of C_{1-1} due to its immediate front vehicle, case 4 is used instead to perform fleet lane changing.

Similarly, when $C_{1-k}, C_{1-(k+1)}, \ldots$, and C_{1-m} have to decelerate to leave S_F, they use their maximum deceleration speeds $a_{1-k}^{\max}, a_{1-(k+1)}^{\max}, \ldots$, and $a_{1-m}^{\max}$ to move outside S_F

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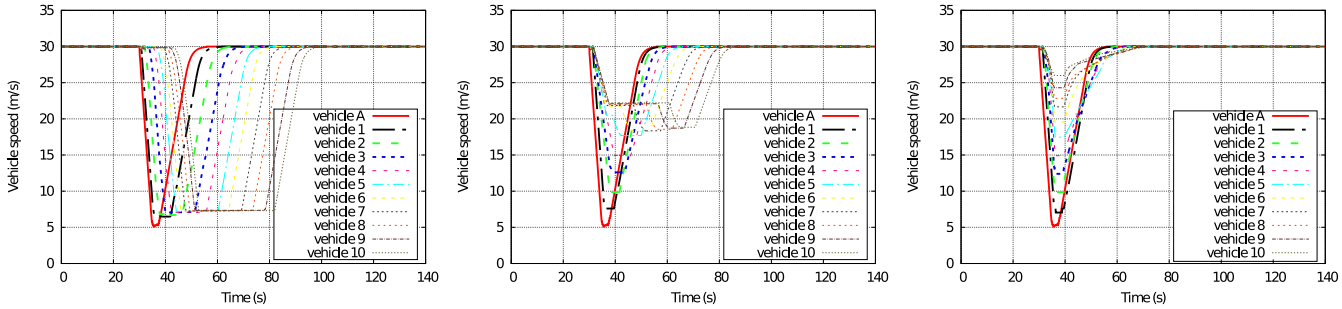

(a) (b) (c)

Fig. 8. Comparisons of vehicle speeds encountering an emergency brake for different fleets of (a) KCFM, (b) TPCC, and (c) CFCC vehicles.

for the periods of t_{1-k}, $t_{1-(k+1)}, \ldots$, and t_{1-m}, respectively. The deceleration periods t_{1-k}, $t_{1-(k+1)}, \ldots$, and t_{1-m} can be estimated as

$$X_{1-k} + E_{1-k} - X_{2-n} = (S_{2-n} - S_{1-k}) \cdot t_{1-k} + \frac{1}{2} \cdot a_{1-k}^{\max} \cdot (t_{1-k})^2 \tag{27}$$

$$X_{1-(k+1)} + E_{1-k} + E_{1-(k+1)} - X_{2-n} = (S_{2-n} - S_{1-(k+1)}) \cdot t_{1-(k+1)} + \frac{1}{2} \cdot a_{1-(k+1)}^{\max} \cdot (t_{1-(k+1)})^2 \tag{28}$$

$$\vdots$$

$$X_{1-m} + \sum_{i=k}^{m} E_{1-i} - X_{2-n} = (S_{2-n} - S_{1-m}) \cdot t_{1-m} + \frac{1}{2} \cdot a_{1-m}^{\max} \cdot (t_{1-m})^2. \tag{29}$$

Case 4: When C_{2-1} has to take a brake before fleet lane changing can be done by F during the deceleration of C_{1-k}, the deceleration speed a_{2-1} and period t_{2-1} of C_{2-1} can be found by solving the following simultaneous equations:

$$\begin{cases} S_{2-1} \cdot T_{\text{emg}} = S_{2-1} \cdot t_{2-1} - \frac{1}{2} \cdot a_{2-1} \cdot (t_{2-1})^2 \\ X_{1-k} + E_{1-k} - X_{2-n} + \frac{1}{2} \cdot a_{2-1} \cdot (t_{2-1})^2 \\ \quad = (S_{2-n} - S_{1-k}) \cdot t_{2-1} + \frac{1}{2} \cdot a_{1-k}^{\max} \cdot (t_{2-1})^2. \end{cases} \tag{30}$$

Note that if C_{1-k} is decelerating to completely stopping before moving outside S_F, the right-hand side of (30) for C_{1-k} has to be replaced by the equation similar to (19).

On the other hand, if all neighboring vehicles C_{1-1}, $C_{1-2}, \ldots$, and C_{1-m} must accelerate to move outside S_F (i.e., $C_{1-(k-1)} = C_{1-m}$), a_{2-1} and t_{2-1} of C_{2-1} can be found by solving the following simultaneous equations:

$$\begin{cases} S_{2-1} \cdot T_{\text{emg}} = S_{2-1} \cdot t_{2-1} - \frac{1}{2} \cdot a_{2-1} \cdot (t_{2-1})^2 \\ X_{2-1} + E_{2-1} - X_{1-m} - \frac{1}{2} \cdot a_{2-1} \cdot (t_{2-1})^2 \\ \quad = (S_{1-m} - S_{2-1}) \cdot t_{2-1} + \frac{1}{2} \cdot a_{1-m}^{\text{acc}} \cdot (t_{2-1})^2. \end{cases} \tag{31}$$

Note that if C_{1-m} is accelerating to the maximum speed limit $S_{\max}$ of the road before moving outside S_F, the right-hand side of (31) for C_{1-m} has to be replaced by the equation similar to (17). In particular, if $a_{2-1} \times t_{2-1}$ is greater than S_{2-1}, F is completely stopping instead of performing lane changing for collision avoidance.

TABLE III
SIMULATION PARAMETERS

Parameter	Value
Fleet Size	$2 \sim 20$ vehicles
Vehicle Speed	$14 \sim 30$ m/s
Maximum Acceleration	6 m/s^2
Maximum Deceleration	8 m/s^2
Driver's Reaction Time	1.5 s
Communication Range	300 m
Lane Width	4 m

IV. PERFORMANCE EVALUATION

The proposed CFCC framework is implemented in a simulator using MATLAB/Simulink [32] with the Krauss car following model (KCFM) [33] and lane-changing duration predictive model [34], where vehicle dynamics are approximated by a first-order system [18], [35]:

$$\tau \dot{a}_i + a_i = a_{i_des} \tag{32}$$

where a_i is the vehicle's actual acceleration, a_{i_des} is the desired acceleration, and τ is a constant time lag. Fleets with 2, 4, ..., and 20 vehicles in the original lane (i.e., lane 2) and different numbers of neighboring vehicles in the target lane (i.e., lane 1) are simulated, where the maximum communication range of each vehicle is 300 m by default. Basic parameters used in our simulation are summarized in Table III. The speeds of fleet vehicles are set from 50 to 110 km/h (i.e., around 14–30 m/s) and their maximum acceleration and deceleration are 6 and 8 m/s^2, respectively. For the recovery time needed to get the intended speed back after the emergency brake of the vehicle ahead of the fleet, our scheme is compared with TPCC [18] and KCFM [33] methods. For the total time needed to change from the original lane to the target lane as necessary, our scheme is compared with SMPC [21] and CDLC [22] methods.

Fig. 8 shows the comparisons of vehicle speed variations using different car following and cruise control methods as the vehicle A ahead of the fleet consisting of vehicles 1, 2, ..., and 10 takes an emergency brake at the 30th second and then starts accelerating (to the original speed) at the 34th second. We can observe that our scheme has the lower braking speed and shorter recovery time than KCFM and TPCC methods for fleet vehicles 2, 3, ..., and 10, as shown in Fig. 8(c). This is because our CFCC uses the extended FFR to

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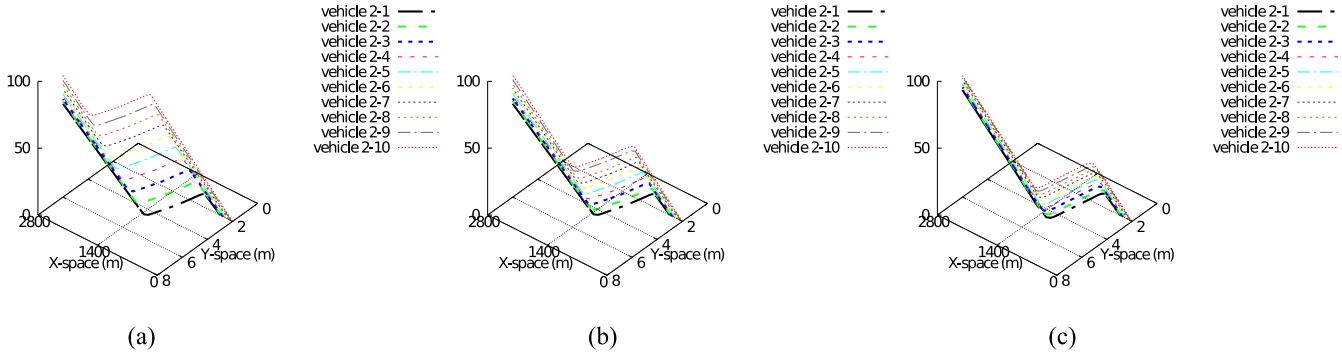

Fig. 9. Comparisons of lane changing trajectories with no neighboring vehicle for different fleets of (a) CDLC, (b) SMPC, and (c) CFCC vehicles.

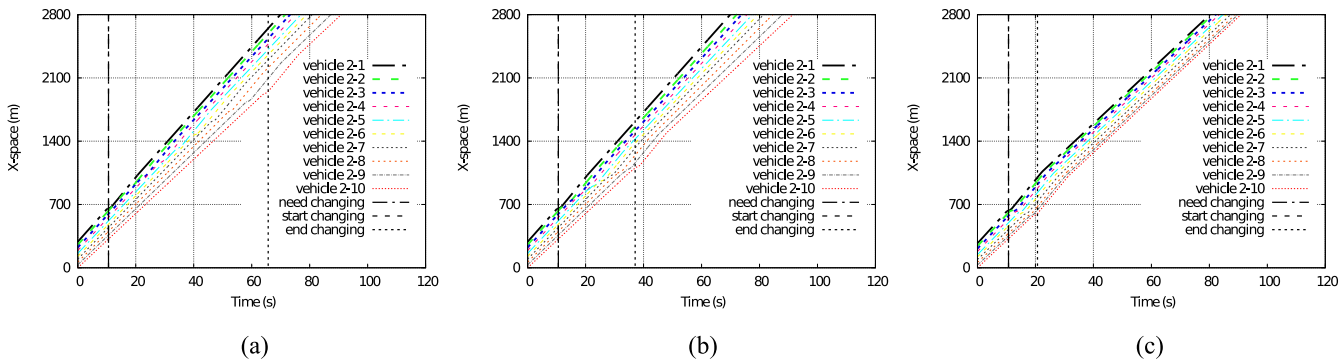

Fig. 10. Comparisons of lane changing times with no neighboring vehicle for different fleets of (a) CDLC, (b) SMPC, and (c) CFCC vehicles.

instantly determine the deceleration/acceleration of remaining CFCC vehicles at the same time when receiving the braking/accelerating messages from the first CFCC vehicle. On the contrary, the KCFM vehicles using the KCFM start braking after seeing the braking lights of their immediate front vehicles, as shown in Fig. 8(a). In addition, the driver's reaction time is needed for each KCFM vehicle for braking and accelerating. On the other hand, the TPCC vehicles using TPCC decide their deceleration based on the DSSM risk [17]. In particular, some following TPCC vehicles brake more than once, where the first braking is caused by the emergency braking of the first TPCC vehicle and the second braking is caused by the consequential braking of the immediate front TPCC vehicle, as shown in Fig. 8(b).

For fleet lane changing, Fig. 9(a)–(c) shows the comparisons of lane changing trajectories (from lane 2 to lane 1) with no neighboring vehicle in lane 1 for different fleets in lane 2 using CDLC, SMPC, and CFCC methods, respectively, where the space between lines could be considered as the distance between vehicles. It can be observed that the fleet of CFCC vehicles has more compact trajectories (i.e., smaller following distances between vehicles) of lane changing than fleets of CDLC and SMPC vehicles. The reason is that the following CFCC vehicles (except the first CFCC vehicle) can change from the original lane to the target lane as soon as possible once detecting that their immediate front CFCC vehicles start changing from the original lane (before completely changing to the target lane). In contrast, the following CDLC vehicles perform lane changing sequentially after their immediate front CDLC vehicles change to the target lane completely (with driver's reaction time), whereas the following SMPC vehicles start changing the lane immediately after the lane changing of their immediate front SMPC vehicles (without driver's reaction time).

Fig. 10 shows the comparisons of lane changing times with no neighboring vehicle in lane 1 for different fleets of CDLC, SMPC, and CFCC vehicles in lane 2, where three dashed lines of need changing, start changing, and end changing indicate the time points at which fleet vehicles $2-1$, $2-2, \ldots$, and $2-10$ need, start, and finish lane changing, respectively. The time line of need changing is overlapped with that of start changing because there is no neighboring vehicle in lane 1 (i.e., $|N| = 0$) that already has the sufficient space for fleet lane changing. All fleets of CDLC, SMPC, and CFCC vehicles start lane changing at the 10th second but finish lane changing at the 67.2th, 39.2th, and 21.3th second, respectively. The reason is similar to what is discussed earlier. Our scheme can reduce the total time of lane changing over existing methods, especially for a long fleet with more following vehicles.

Fig. 11 shows the comparisons of lane changing trajectories with one neighboring vehicle (i.e., vehicle $1-1$) in lane 1 for different fleets of CDLC, SMPC, and CFCC vehicles in lane 2. Similarly, the CFCC fleet has the most compact trajectories of lane changing among all methods. Vehicle $2-1$ sends the request message of lane changing to vehicle $1-1$ and then changes from lane 2 to lane 1 as vehicle $1-1$ accelerates to move outside the needed lane changing space in lane 1. Fig. 12 shows the comparisons of lane changing times with one neighboring vehicle in lane 1 (i.e., $|N| = 1$) for different fleets of CDLC, SMPC, and CFCC vehicles in lane 2. All fleets

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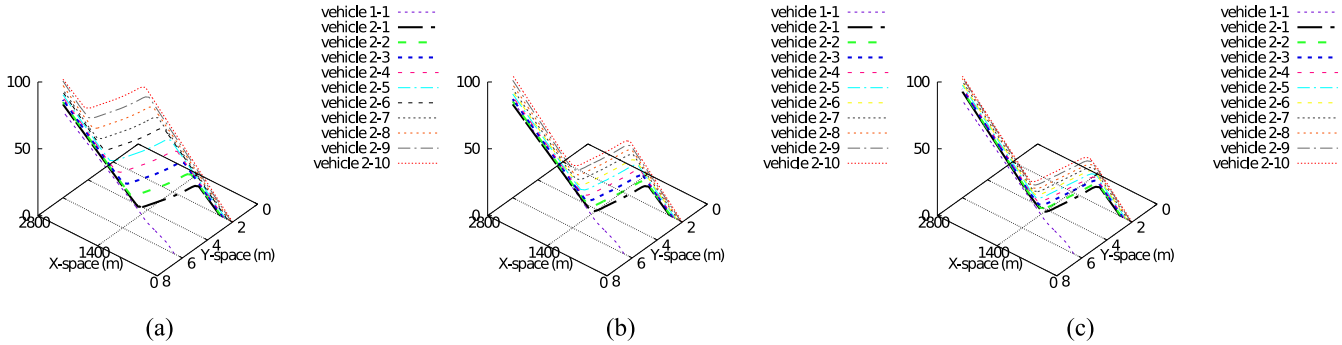

Fig. 11. Comparisons of lane changing trajectories with one neighboring vehicle for different fleets of (a) CDLC, (b) SMPC, and (c) CFCC vehicles.

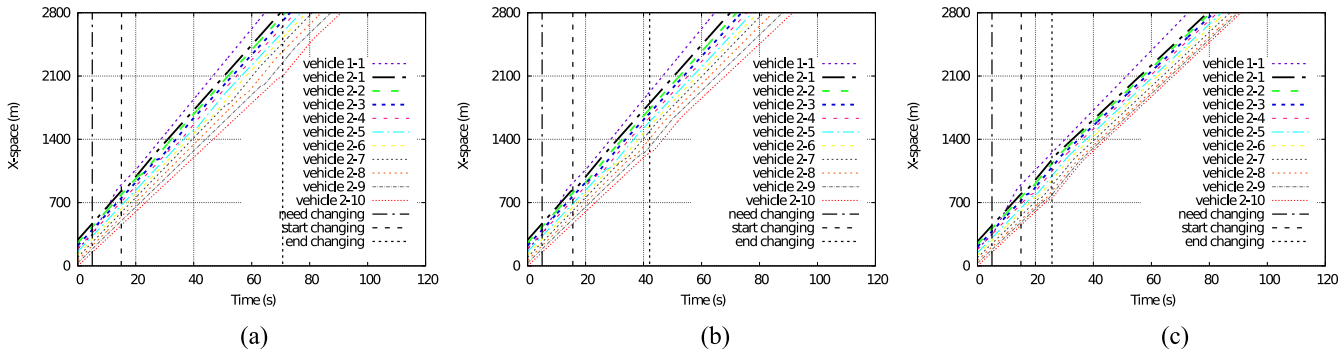

Fig. 12. Comparisons of lane changing times with one neighboring vehicle for different fleets of (a) CDLC, (b) SMPC, and (c) CFCC vehicles.

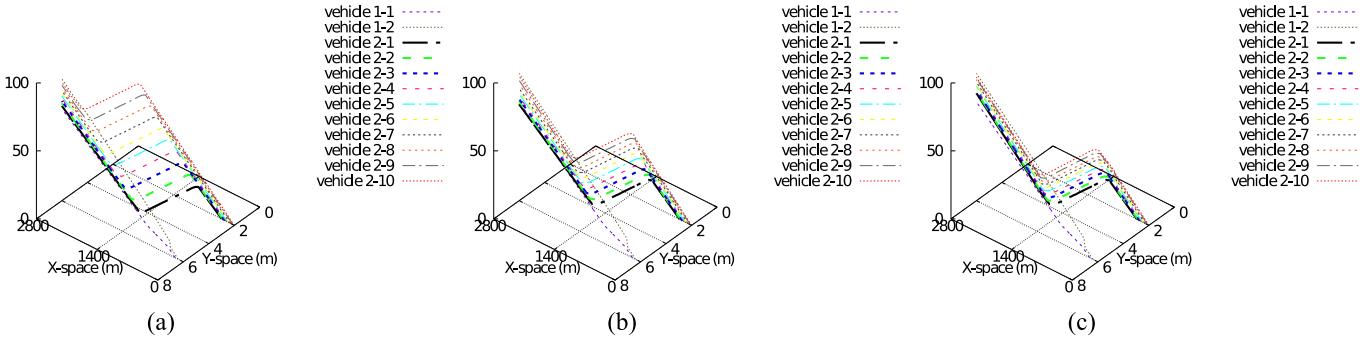

Fig. 13. Comparisons of lane changing trajectories with two neighboring vehicles for different fleets of (a) CDLC, (b) SMPC, and (c) CFCC vehicles.

of CDLC, SMPC, and CFCC vehicles need lane changing at the 5th second but finish lane changing at the 74.3th, 42.4th, and 28.2th second, respectively. Our scheme can offer the most efficient lange changing with the shortest total time, where the reason is similar to that for Fig. 10.

Fig. 13 shows the comparisons of lane changing trajectories with two neighboring vehicles (i.e., vehicles $1-1$ and $1-2$) in lane 1 for different fleets of CDLC, SMPC, and CFCC vehicles in lane 2. Similarly, vehicle $2-1$ sends the request message of lane changing to vehicles $1-1$ and $1-2$ and then changes from lane 2 to lane 1 as vehicles $1-1$ and $1-2$ accelerates and decelerates to move outside the needed lane changing space in lane 1, respectively. Fig. 14 shows the comparisons of lane changing times with two neighboring vehicles in lane 1 (i.e., $|N| = 2$) for different fleets of CDLC, SMPC, and CFCC vehicles in lane 2. All CDLC, SMPC, and CFCC fleets need lane changing at the 5th second but finish lane changing at the 73.1th, 42.6th, and 27.7th second, respectively. It can be seen that the lane changing time of CDLC vehicles is sightly shorter than that with only one neighboring vehicle in lane 1. This is because the CDLC vehicles change the lane individually without waiting until the entire space of fleet lane changing is available. In contrast, the SMPC and CFCC vehicles have to wait for the sufficient space of lane changing for the entire fleet with ten vehicles.

Fig. 15 shows the comparisons of lane changing trajectories with two neighboring vehicles in lane 1 and fleet vehicles with braking in lane 2 for different fleets of CDLC, SMPC, and CFCC vehicles. Different from Fig. 13, vehicle $2-1$ sends the request message of lane changing to vehicles $1-1$ and $1-2$ and starts braking until vehicles $1-1$ and $1-2$ accelerates to move outside the needed lane changing space in lane 1. Fig. 16 shows the comparisons of lane changing times with two neighboring vehicles in lane 1 and braking fleet vehicles in lane 2

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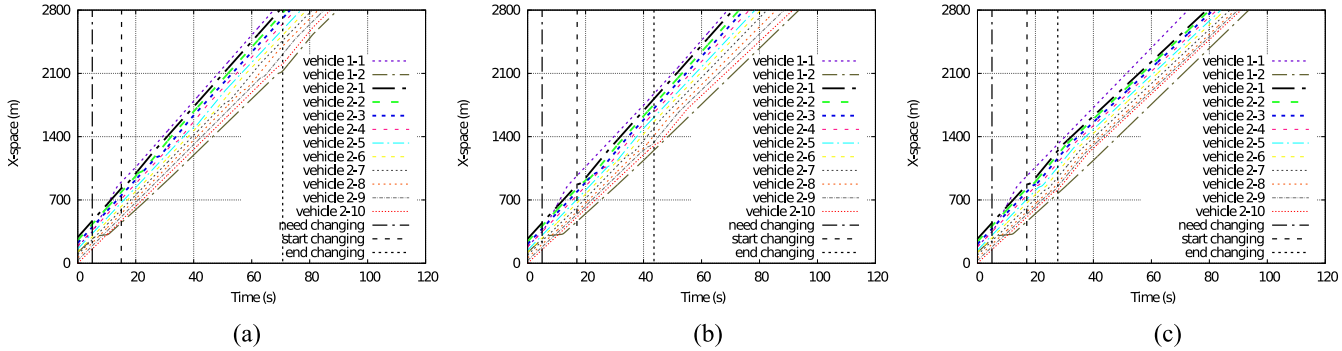

Fig. 14. Comparisons of lane changing times with two neighboring vehicles for different fleets of (a) CDLC, (b) SMPC, and (c) CFCC vehicles.

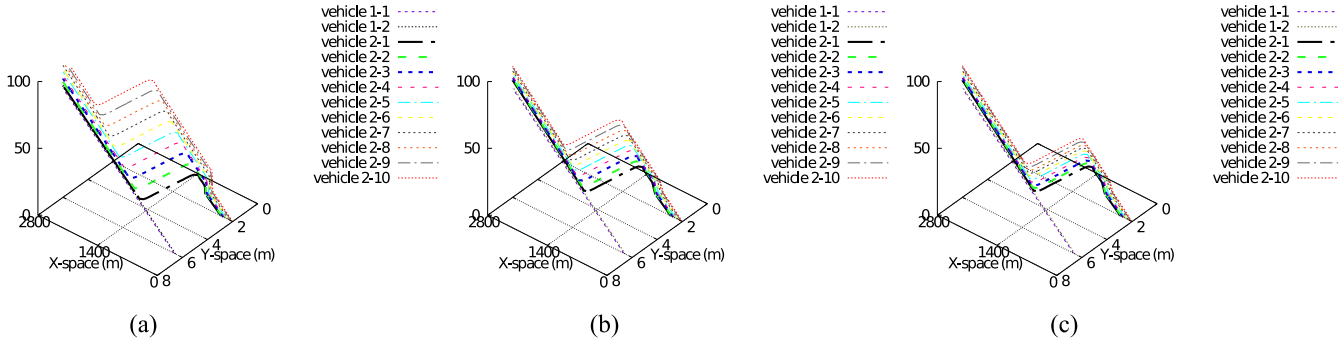

Fig. 15. Comparisons of lane changing trajectories with two neighboring vehicles and braking fleet vehicles for different fleets of (a) CDLC, (b) SMPC, and (c) CFCC vehicles.

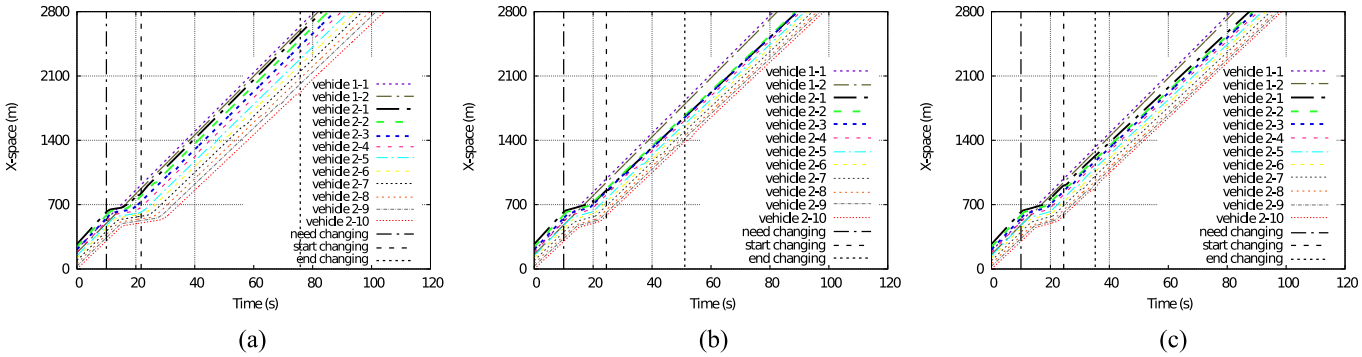

Fig. 16. Comparisons of lane changing times with two neighboring vehicles and braking fleet vehicles for different fleets of (a) CDLC, (b) SMPC, and (c) CFCC vehicles.

for different fleets of CDLC, SMPC, and CFCC vehicles. All of CDLC, SMPC, and CFCC vehicles need lane changing at the 10th second but start lane changing at the 21st, 25th, and 25th second and finish lane changing at the 78.5th, 54.2th, and 36.8th second, respectively. It can be seen that although the first CDLC vehicle starts lane changing at the earliest time point, the last CDLC vehicle finishes lane changing at the latest time point among all methods due to individual lane changing. In particular, the CDLC fleet does not determine the acceleration periods for neighboring vehicles. In contrast, the SMPC and CFCC vehicles start lane changing at the later time points but finish lane changing at the earlier time points due to fleet lane changing. In particular, the comparisons of total lane changing times (between needing and finishing lane changing) of CDLC, SMPC, and CFCC fleets in lane 2 with 0, 1, and 2 neighboring vehicles in lane 1 are summarized in Table IV.

TABLE IV
TOTAL LANE CHANGING TIME

Case	$\|N\| = 0$	$\|N\| = 1$	$\|N\| = 2$	$\|N\| = 2$ with fleet braking
CDLC	57.2 sec	69.3 sec	68.1 sec	68.5 sec
SMPC	29.2 sec	37.4 sec	37.6 sec	44.2 sec
CFCC	11.3 sec	23.2 sec	22.7 sec	26.8 sec

Next, we further compare total lane changing times for different numbers of CDLC, SMPC, and CFCC vehicles in a fleet with no neighboring vehicle, as shown in Fig. 17. Note that the results with one or more neighboring vehicles are similar to those with no neighboring vehicle. It can be observed

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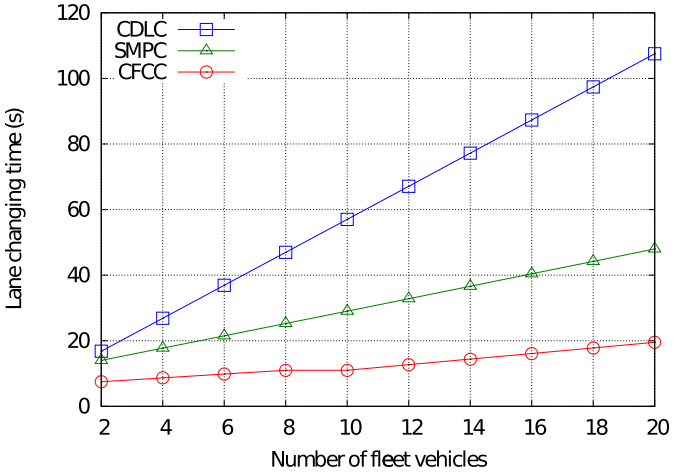

Fig. 17. Comparisons of total lane changing times for different numbers of (a) CDLC, (b) SMPC, and (c) CFCC vehicles in a fleet.

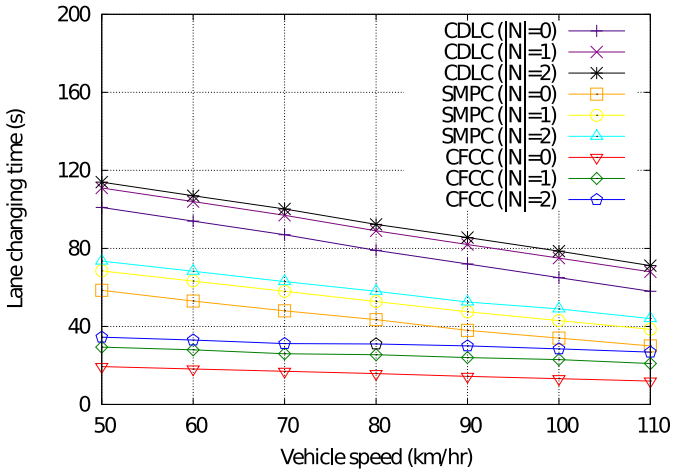

Fig. 18. Comparisons of total lane changing times under different vehicle speeds for 10-vehicle (a) CDLC, (b) SMPC, and (c) CFCC fleets.

that our scheme has the shortest total lane changing time for all fleets with 2, 4, . . ., and 20 vehicles among all methods. In addition, for more vehicles in the fleet, our scheme can reduce more lane changing time than other methods. On the other hand, Fig. 18 shows the comparisons of total lane changing times under different vehicle speeds for 10-vehicle CDLC, SMPC, and CFCC fleets with 0, 1, and 2 neighboring vehicles. Similarly, our scheme has the shortest total lane changing time for all fleets driving at 50, 60, . . ., and 110 km/h among all methods, which can cooperatively provide the sufficient space for lane changing and efficiently perform lane changing for fleets with various speeds as fast as possible.

V. Conclusion

The CFCC framework is designed for collision avoidance and lane changing based on VSNs. For traffic safety, the designed framework estimates the CFR of a single vehicle and extends the estimated risk to the FFR of multiple vehicles. For driving efficiency, our framework uses the extended FFR to coordinate the velocity control of fleet vehicles for reducing the recovery time to get the original speed back after sudden braking. For lane changing, the framework determines the necessary acceleration and deceleration of individual vehicles in the target lane for providing the sufficient space to fleet vehicles. In addition, the feasible deceleration of fleet vehicles are cooperatively determined for lane changing when fleet braking is necessary. For future works, we are exploring the automatic splitting and merging of fleets with precise vehicle dynamics and lateral control stability in our framework to achieve more efficient and flexible lane changing based on accurate positioning information using Internet-of-Vehicles technologies.

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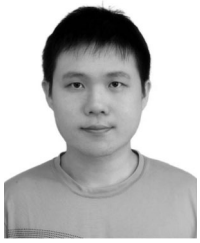

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